Introduction to Coding theory

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We cover the following topics in this note.

- TBA.

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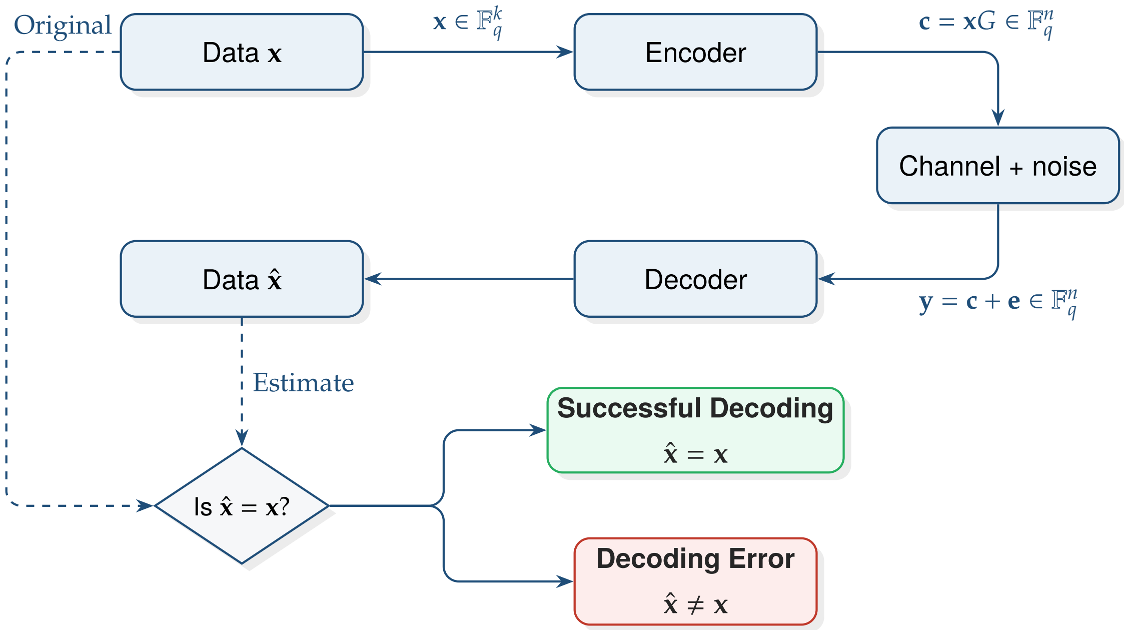


A generic communication model has the form

$$\mathbf{x} = (x_1, \ldots, x_k) \xrightarrow{\text{encoder}} \underbrace{\mathbf{c} = (c_1, \ldots, c_n)}_{\text{codeword } (n \geq k)} \xrightarrow{\text{noise } \mathbf{e}} \mathbf{y} = \mathbf{c} + \mathbf{e} \xrightarrow{\text{decoder}} \hat{\mathbf{x}} = (\hat{x}_1, \ldots, \hat{x}_k).$$

Example 1 (Repetition code)**.** Let Σ be an alphabet. An *encoding* is a map

$$\begin{array}{rrcl} \text{Encode} : & \Sigma^k & \longrightarrow & \Sigma^n \\ & (x_1, \dots, x_k) & \longmapsto & (c_1, \dots, c_n) \end{array}$$

with $n \geq k$. When $k = 1$, a very simple encoding repeats the same symbol

$$x_1 \xmapsto{\text{Encode}} \underbrace{(x_1,\, x_1,\, \dots,\, x_1)}_{n \text{ copies } (\in \Sigma^n)}$$

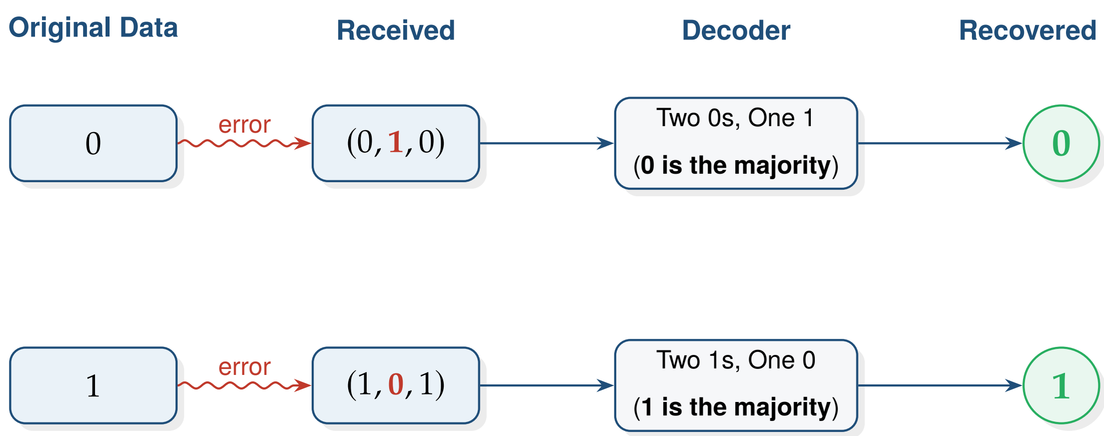
$$0 \longmapsto (0,\, 0,\, \dots,\, 0)$$

$$1 \longmapsto (1,\, 1,\, \dots,\, 1)$$

This is called the *repetition code*. If one symbol is changed during transmission, we can still recover the original message by *majority vote*. For instance,

$$(0,0,0) \rightsquigarrow (0,1,0), \qquad (1,1,1) \rightsquigarrow (1,0,1).$$

Since 0 appears most often in $(0,1,0)$, we decode it as 0, and since 1 appears most often in $(1,0,1)$, we decode it as 1.



2.1 Linear Codes

Linear Encoding

Definition 1. A **linear encoding** over $\mathbb{F}_q$ is a linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{x} \longmapsto \mathbf{c}.$$

Remark 1. Since the map is linear,

- the sum of two codewords is again a codeword;
- any scalar multiple of a codeword is again a codeword.

Linear Code

Definition 2. A linear subspace $C \subseteq \mathbb{F}_q^n$ is called a **linear code**.

If C is a linear code, then automatically:

- the zero codeword $\mathbf{0}$ belongs to C;
- whenever $\mathbf{c} \in C$, also $-\mathbf{c} \in C$;
- every codeword is a linear combination of basis vectors of C.

Proposition 1. *If C is an (n, k) linear code over $\mathbb{F}_q$, then*

$$|C| = q^k.$$

Proof. Choose a basis $\mathbf{b}_1, \ldots, \mathbf{b}_k$ of C. Every codeword can be written uniquely as

$$x_1\mathbf{b}_1 + \cdots + x_k\mathbf{b}_k, \qquad x_i \in \mathbb{F}_q.$$

Each coefficient x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k. □

A Exercises

1. Consider a communication channel, which takes as input a vector of length $n = 5$, and erases exactly one out of the 5 coefficients, but we do not know which one. Design two linear codes over $\mathbb{F}_3$ that make sure any message of length $k = 4$ can always be received perfectly at the receiver.

Sol. Consider a communication channel that takes a vector of length $n = 5$ and erases exactly one coefficient. We want to design two linear codes over $\mathbb{F}_3$ that allow a message of length $k = 4$ to be recovered perfectly.

Theory

A linear code can correct e erasures if and only if its minimum distance d satisfies $d \geq e + 1$. To correct $e = 1$ erasure, we need a code with $d \geq 2$. In terms of the parity-check matrix H, $d \geq 2$ is achieved if and only if no column of H is the zero vector.

Code 1: Single Parity-Check Code

The simplest approach is to add a parity digit that is the negative sum of the message digits, ensuring the sum of all components is zero in $\mathbb{F}_3$.

Generator Matrix G_1

For a systematic code where the first 4 symbols are the message, we define:

$$G_1 = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Parity Check Matrix H_1

The parity check matrix H_1 satisfies $G_1 H_1^T = 0$. In $\mathbb{F}_3$:

$$H_1 = \begin{pmatrix} 2 & 2 & 2 & 2 & 1 \end{pmatrix}$$

Note: Any non-zero multiple of this row also works. Since every entry is non-zero, any single erased symbol can be uniquely determined from the remaining four.

Code 2: Weighted Parity-Check Code

We can use different coefficients to ensure the code is distinct. We assign weights (1 and 2) to the message symbols.

Generator Matrix G_2

$$G_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

Parity Check Matrix H_2

The resulting check equation for the parity bit $x_5 = x_1 + 2x_2 + x_3 + 2x_4$ gives:

$$H_2 = \begin{pmatrix} 2 & 1 & 2 & 1 & 1 \end{pmatrix}$$

Conclusion

In both designs, no column of the parity-check matrix is zero. Therefore, $d \geq 2$. If any one symbol x_i is erased, the receiver can solve the single linear equation defined by $H \cdot \mathbf{c}^T = 0$ for the missing variable x_i because the i-th coefficient in the parity check is non-zero in $\mathbb{F}_3$. □

References

[1] F. Oggier, *Teaching Material: Coding Theory*, feog.github.io. Available: `https://feog.github.io/`. Accessed: 2026-04-01.

[2] F. Oggier, *Coding Theory: Introduction*. Available: `https://feog.github.io/1-coding.pdf`. Accessed: 2026-04-01.

[3] F. Oggier, *Coding Theory: Linear Codes*. Available: `https://feog.github.io/2-coding.pdf`. Accessed: 2026-04-01.